An Analysis of Bristol's Flood Exposure Using QGIS

Sara Sari · November 2026

Introduction

Flood events across the United Kingdom are occurring more often with greater severity and the number of people exposed to frequent flooding is expected to rise by 61% by 2050 and 118% by 2080. Furthermore, 6.9 million properties are currently at risk of flooding, but only 39% of people know how to act if they receive a flood warning.[1]

This risk is particularly acute in Bristol, which has experienced over 20 tidal floods in the past decade and in 2020 recorded its highest tide ever.[2] To understand which parts of the city are exposed, I mapped Bristol's 54,726 buildings against the Environment Agency's (EA) 2025 flood zones using QGIS.[3]

Findings

Figure 1 shows that 2,319 buildings (4.2% of Bristol's buildings) lie in Flood Zone 3, the highest-risk zone, with a 1% or greater annual chance of fluvial flooding, or 0.5% or greater of tidal flooding.[4] The risk is concentrated in two areas, namely Avonmouth and the city centre (Figure 1b).

Splitting Bristol into its neighbourhoods (Figures 2 and 2b), the most exposed neighbourhoods by percentage of buildings are around the city centre and Floating Harbour, likely reflecting dense riverside development. For example, Bristol 054F has 95.8% of its buildings in Zone 3, but is in decile 7, among the less deprived half of England's neighbourhoods. By number of buildings, the picture changes as four of the top five neighbourhoods are among the 30% most deprived in England.[5] Additionally, three Avonmouth neighbourhoods alone contain 780 buildings in Zone 3, about a third of Bristol's total.

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[1] British Red Cross, "Flood Risk in the UK: 6.9 Million Homes at Risk of Flooding but Awareness Is Low," n.d., accessed November 2, 2026, https://www.redcross.org.uk/stories/disasters-and-emergencies/uk/what-are-floods.

[2] Bristol City Council, "The Risk," Avon Riversides 2100, n.d., accessed November 2, 2026, https://avonriversides.co.uk/the-risk/.

[3] Environment Agency, *Flood Map for Planning – Flood Zones* (dataset), 2025, accessed October 27, 2026, https://environment.data.gov.uk/dataset/04532375-a198-476e-985e-0579a0a11b47; Ordnance Survey, *OS OpenMap – Local* (dataset), 2026, accessed October 27, 2026, https://www.ordnancesurvey.co.uk/products/os-open-map-local; QGIS Development Team, *QGIS Geographic Information System*, version 3.44, 2025, https://qgis.org/.

[4] Ministry of Housing, Communities and Local Government, "Flood Risk and Coastal Change," Planning Practice Guidance, 2022, para. 077, accessed November 2, 2026, https://www.gov.uk/guidance/flood-risk-and-coastal-change#para77.

[5] Ministry of Housing, Communities and Local Government, *English Indices of Deprivation 2025*, File 1: Index of Multiple Deprivation (dataset), 2025, accessed October 27, 2026, https://www.gov.uk/government/statistics/english-indices-of-deprivation-2025.

With climate change (Figures 3 and 3b), a further 1,059 buildings could be at risk, an increase of 24%, mainly around Avonmouth, Lawrence Weston and the city centre. This is consistent with the council's view that tidal flooding along the Avon will increase with rising sea levels.[6]

The identified areas at risk agree with those highlighted by Bristol City Council. These include Coronation Road and Cattle Market Road, which are predominantly at risk from tidal flooding.[7] My figures also show clusters along Bristol's smaller rivers, such as the Frome at Eastville, reflecting the council's findings of fluvial flooding around Henbury and at the base of the Dundry Hills.[8]

Industry Flood Schemes

I compared my maps and analysis with two major flood schemes in Bristol. These are the Avonmouth Severnside Enterprise Area (ASEA)[9] flood defences and the Bristol Avon Flood Strategy (now Avon Riversides 2100).[10]

For the ASEA project, Mott MacDonald provided numerous services, including coastal and fluvial modelling, to construct flood defences along the Severn Estuary.[11] My finding that Avonmouth has the largest concentration of buildings at risk (Figure 2) matches where these defences have been built.

My analysis also aligns with the Bristol Avon Flood Strategy, the long-term plan for managing flood risk from the River Avon, developed by Bristol City Council with Arcadis, Arup and Mott MacDonald.[12] The proposed defences extend along Hotwells and Cumberland Basin, through the city centre, and at Shirehampton and Sea Mills.[13] These defence lines run alongside the at-risk buildings in Figure 1, particularly at the harbourside, Spike Island, St Philip's Marsh and St Anne's, helping to separate tidal

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[6] Bristol City Council, "The Risk."
[7] Ibid.
[8] Bristol City Council, *Bristol Local Flood Risk Management Strategy* (Bristol: Bristol City Council, 2023), 15, https://democracy.bristol.gov.uk/documents/s82208/Bristol%20Local%20Flood%20Risk%20Management%20Strategy%202023.pdf.
[9] Mott MacDonald, "Avonmouth Severnside Enterprise Area, UK," n.d., accessed November 2, 2026, https://www.mottmac.com/en/projects/avonmouth-severnside-enterprise-area-uk/.
[10] Ove Arup & Partners Ltd, *Bristol Avon Flood Strategy: Options Development Report*, Appendix C: Preferred Option Report (Bristol: Bristol City Council and Environment Agency, 2023), https://www.bristol.gov.uk/files/ask-bristol/9506-appendix-c-preferred-option-report/file.
[11] Mott MacDonald, "Avonmouth Severnside Enterprise Area."
[12] Ove Arup & Partners Ltd, *Options Development Report*.
[13] Ove Arup & Partners Ltd, *Bristol Flood Strategy: Strategy Area (City) – Proposed Defences*, Drawing 391. (Bristol: Bristol City Council, 2020), https://www.bristol.gov.uk/files/ask-bristol/9517-bristol-avon-flood-strategy-appendix-a-key-plans/file.

water from these areas. Furthermore, Arup's 2130 flood extent in the Outline Business Case closely resembles my climate change projection (Figure 3b).[14]

These results align more closely with the Bristol Avon Flood Strategy than with the council's local flood risk figures, as both my data and the strategy's combine tidal and fluvial flooding. Overall, both schemes target the same hotspots that my analysis identifies, even though I used building footprints rather than individual properties. However, clusters of at-risk buildings along smaller rivers, such as those at Eastville (River Frome) and Ashton Vale (Colliter's Brook), lie outside both of these projects (Figure 1), which the strategy itself acknowledges.[15]

Limitations and Next Steps

I acknowledge that the data I used from the EA does not include the flooding risks attributed to surface water, groundwater or sewers. Therefore, my figures are likely to underestimate total flood risk. Conversely, flood zones show potential flooding without existing defences, so actual risk may be lower where defences are in place. Finally, OS OpenMap – Local merges joined-up buildings and includes small structures such as sheds, so 54,726 is a count of building footprints rather than individual properties. Future work could include surface water flood data to give a fuller depiction of flood risk and highlight critical infrastructure at risk, such as hospitals and schools.

Conclusion

Overall, the analysis found that 4.2% of Bristol's buildings lie in the highest flood risk zone, with Avonmouth and the city centre most affected. The majority of the neighbourhoods with the most at-risk buildings are among the 30% most deprived in England, meaning these communities may be less able to recover from flooding. Whilst existing flood schemes cover the primary hotspots, there are some areas along smaller rivers that remain outside of them. With climate change expected to bring about a 24% increase in buildings at risk, this highlights the importance of mapping flood zones and understanding, in particular, the effect of tidal flooding on Bristol.

[14] Ove Arup & Partners Ltd, *Bristol Avon Flood Strategy: Outline Business Case*, P05 (Bristol: Bristol City Council, 2024), 4, fig. 3, https://avonriversides.co.uk/wp-content/uploads/2026/04/Bristol_Avon_Flood_Strategy_OBC_small.pdf.
[15] Ibid., 37.

Figures

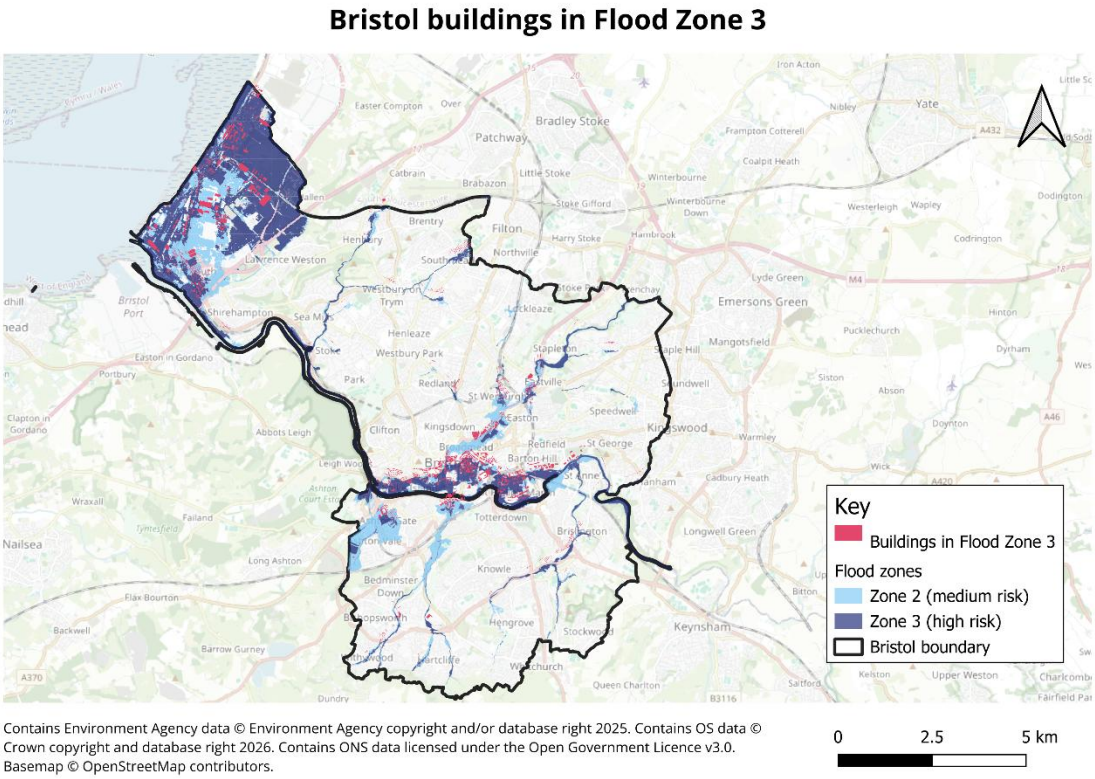


Figure 1: *Buildings in Flood Zone 3, Bristol.*

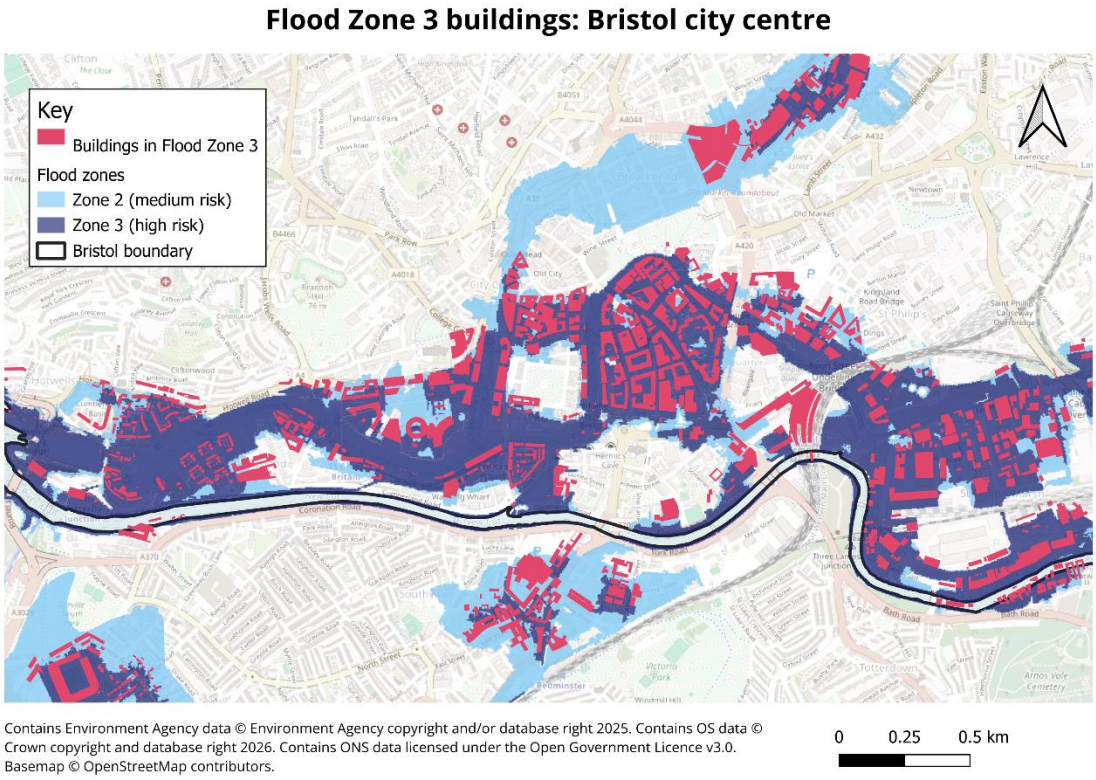


Figure 1b: *Buildings in Flood Zone 3, focused on Bristol city centre.*

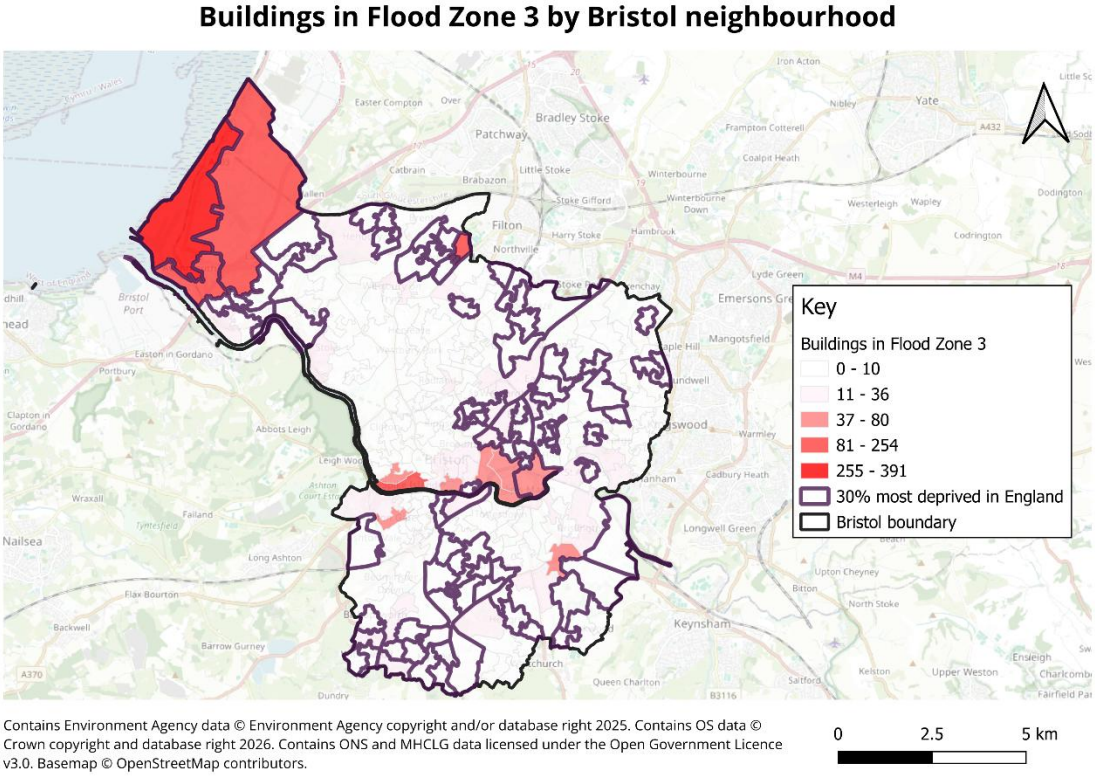


Figure 2: *Buildings in Flood Zone 3 by neighbourhood, with the 30% most deprived neighbourhoods outlined.*

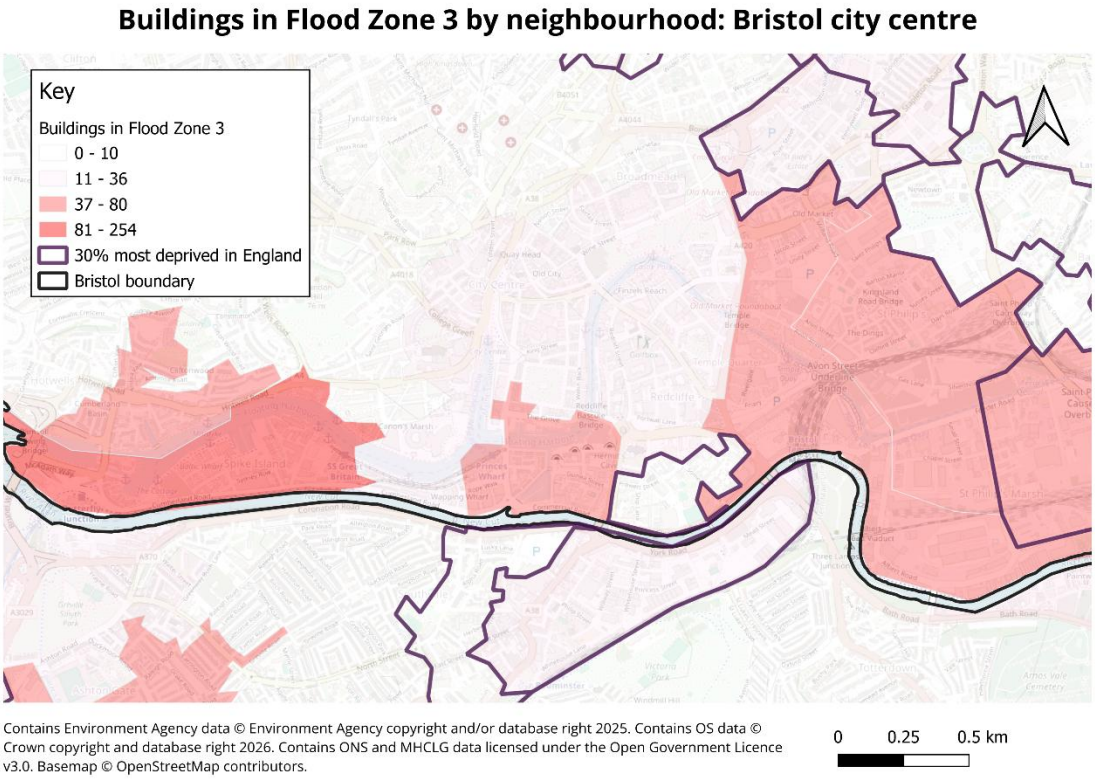


Figure 2b: *Buildings in Flood Zone 3 by neighbourhood, focused on Bristol city centre.*

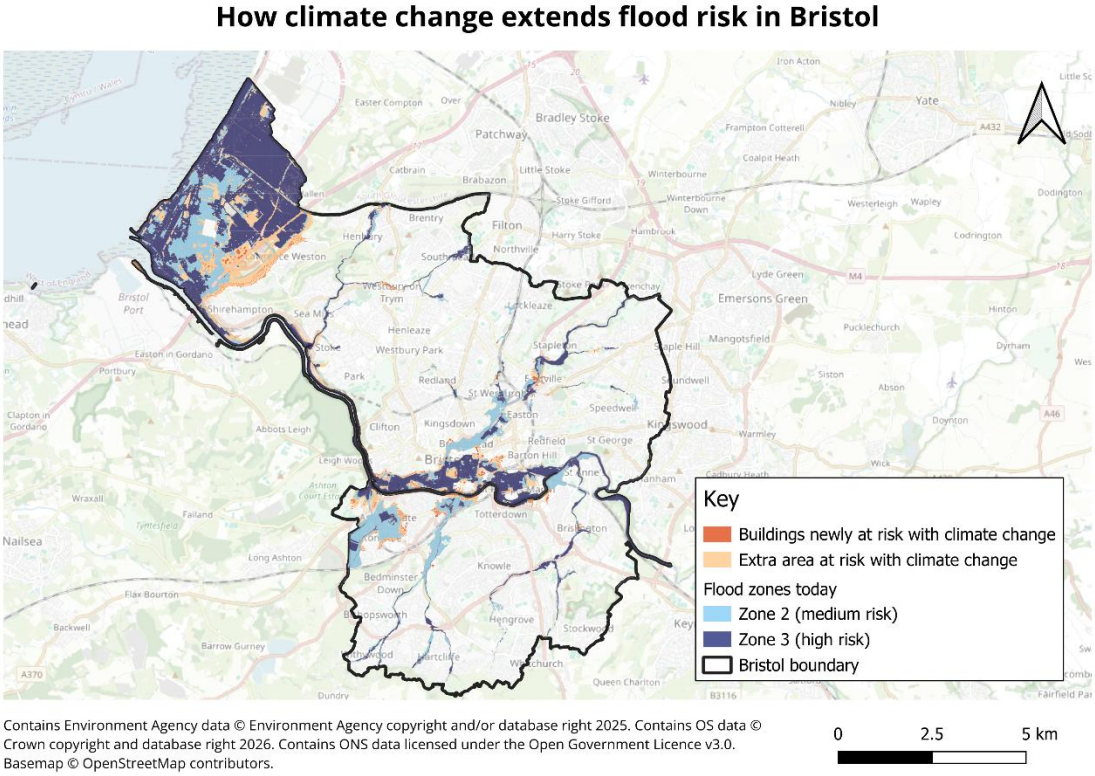


Figure 3: *Additional flood risk with climate change, Bristol.*

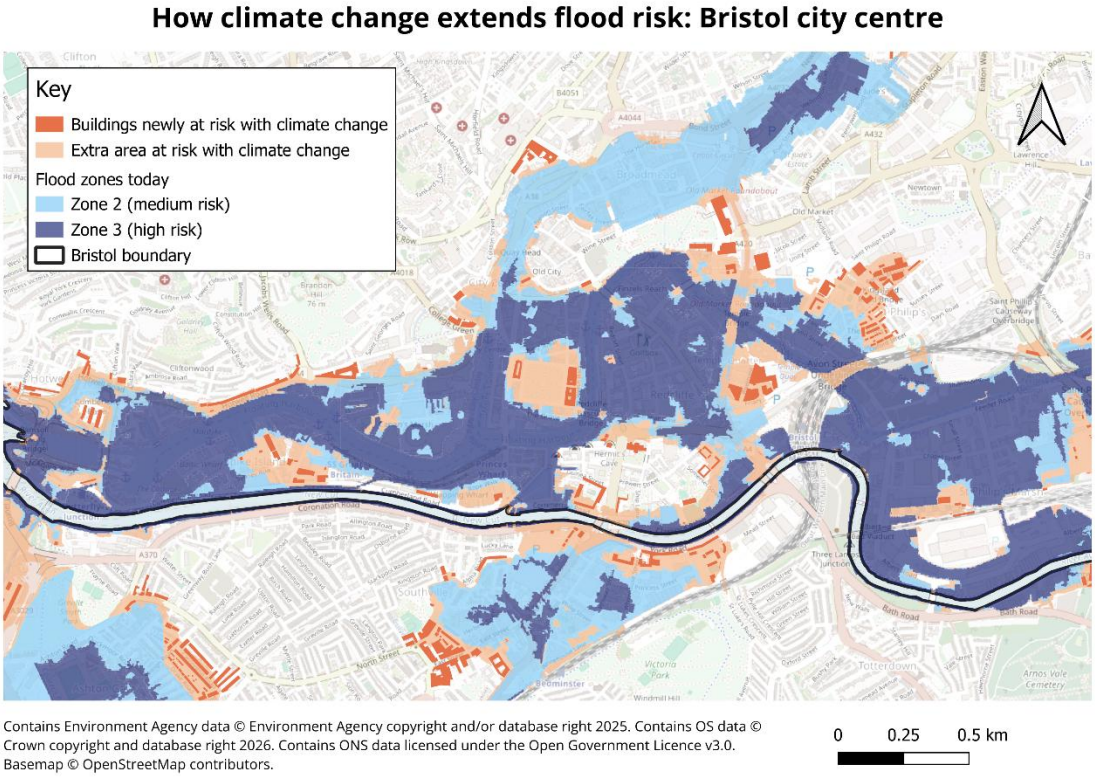


Figure 3b: *Additional flood risk with climate change, focused on Bristol city centre.*

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